THE UNIVERSITY OF BRITISH COLUMBIA

CONFINED DIFFUSION OF NANOPARTICLES USING iSCAT MICROSCOPY

by

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Abstract

Particle confinement allows extended measurements of suspended biomolecules and nanoparticles using optical microscopy as it keeps it within the field of view.

However, during confinement, the size-diffusivity relationship changes from that in bulk, where the correction factor is typically assumed without consideration of the occurring measurement conditions such as surface charges and the media.

In this work, we show that by combining interferometric scattering (iSCAT) microscopy with microfluidic nanoparticle confinement we can directly measure the positional distribution and depth-dependent diffusivity. We demonstrate how it depends on particle charge and media properties for both reference beads and lipid nanoparticles. Utilizing range-of-depth motion and in-plane diffusivity as complementary diameter estimates, we showcase how 3D particle tracking under confinement allows for nanoparticle diameter quantification in protein-containing media.

The ability to measure the effect of confined diffusivity improves accuracy in trajectory-based particle sizing in varying media, commonly found in cell biology and nanoparticle sizing.[1].

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Chapter 1

Introduction

1.1 Movivation

- What is the major field i am in? - what are the biological questions - how are they typically slved

- problems with the current solutions

In much of nature, the idea of size scales is widely discussed and investigated. From a basic understanding of physics, one understands that the forces an object feels are directly linked to their mass, or volume and mass density. For example, we can easily understand what will happen to a basketball if we throw it up into the air, as we are sitting on planet earth. However, when we leave the familar size scale, such as having a basketball the size of planet earth, or the size of a gas molecule, our intuition begins to show its cracks.

In size regime of nanoparticles, that is between 20-100nm, many biological particles of utmost importance tend to reside. For instance, in cellular biology, proteins, vesicles, enzymes, and organelles fit between 4 nm to 10 µm. In this size regime, the forces which one feels largely differ from that of say a basketball. Inter-molecular forces between nearby molecules have a strong effect on particle postion, thermal kicks from atomic molecules induce a stochastic motion non-determininstically, and forces of hydrodynamic pressure induce frictions slowing the particle down. Thus, to have any quantitative understnandig we must try to

1.2 Outline for Thesis

- Chapter 1 is an introduction/background - Chpater 2 discusses application of Widefield iSCAT for SPT in the Rayleigh Regime - Conclusion

Chapter 2

SPT in confinement principles

Biology is mostly nanoparticles. Nanoparticles move around stochastically under the rules of brownian motion. To be able to measure the particle's characteristics, one can relate their motion and stochasticity to their properties. This chapter will discuss the phenomena of Brownian motion and how it relates to particle properties, environmental ion concentration, viscosity, hydrodynamic boundaries, and confinement. To directly measure particle position, one can use optical methods, specifically, optical scattering measurements. In this chapter, the theoretical background regarding the particle's stochastic motion is presented, with particular focus on how the processes evolve in confined geometries. There will be a discussion on relating optical signals to their motion, as well as optical methods for particle detection.

Methods that are able to detect single nanoparticles are known as single-molecule measurements. In the field of optical microscopy, single-molecule methods are often referred to as single particle tracking (SPT). This chapter aims to direct the reader through the theoretical context of the work done. First, Brownian motion is discussed, then optical preliminaries for SPT, and lastly, biological motivation is touched upon to continue the introductory words.

2.1 Brownian Motion

- Introduce the concept of Brownian Motion - Histroy of brownian Motion - applications of simple

- discuss an outlook of what has been done since then - outline of this section

2.1.1 Langevin Dynamics

- Brownian motion can alternatively be described using a Langevin Model. - introduce langevin equation - velocity correlation - Arrive at the Fluc-Dissipation - relate it to Einstein's formulation and show that given a stokes drag (sphere)

$$D = \frac{k_b T}{\gamma_0} \tag{2.1}$$

2.1.2 Smoluchowski Equation

- Since Langevin is kind of convient, how do we acctualy link the two methods?

- Fokker plank introdution motivated from the master equation - - ideally summarize van kampen.

- arrive at the general Smoluchowski equation for a random walker - simple plots for descriptive purposes.

2.2 Confined Diffusion Processes

Although diffusion is interesting, there is a wealth of complexity in confinement due to Hydrodynamic effects - quote 111 years here

- outline of this seciton

2.2.1 Hydrodynamic Confinement

- confinemnt from Hydrodynamics - frictional slowing - ...

2.2.2 Geometric Confinement

- i need to check references on if this is relate

2.2.3 Electrostatic Potentials

2.2.4 Example: Polystyrene Bead

2.3 In-Vivo Considerations

- Now that we more or less know how BM works, why is it important in cells?

2.3.1 why does this matter for biological applications

2.3.2 Examples of Brownian Motion in Biology

2.4 Brownian motion as a method for measurements of size in biology

2.4.1 survey of methods using in this field

- Optical Methods - Bulk: DLS, - Single Molecule: SPT, NTA, FCS,

2.4.2 how confinemen effects these measurements

- Literature on why these methods are lacking

2.4.3 Probing of surface interactions with these methods

Chapter 3

Widefield backscattering iscat for SPT in 3d confinement tracking and analysis

3.1 Questions

3.2 Rayleigh scattering of nanoparticles

3.2.1 Parameterization used in this thesis

3.3 Methods

3.3.1 Optical System

3.3.2 Convex-lens induced confinement

3.4 Results i guess

3.5 Discussion of Results

Chapter 4

Conclusion

4.1 Conclusion

4.2 Future outlook

Bibliography

[1] Albert Kamanzi et al. "Simultaneous, Single-Particle Measurements of Size and Loading Give Insights into the Structure of Drug-Delivery Nanoparticles". en. In: *ACS Nano* 15.12 (Dec. 2021), pp. 19244–19255. ISSN: 1936-0851, 1936-086X. DOI: 10.1021/acsnano.1c04862. URL: https://pubs.acs.org/doi/10.1021/acsnano.1c04862 (visited on 06/18/2026).